CBSE Test Paper 05

Chapter 11 Thermal Properties of Matter

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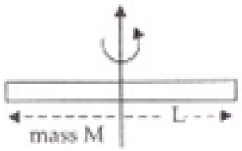
1. A brass spring has a spring constant k. When the spring is heated, the spring constant will **1**
   a. remain the same
   b. increase
   c. increase, then decrease
   d. decrease
2. In an insulated vessel, 250 g of ice at 0^oC is added to 600 g of water at 18.0^oC. What is the final temperature of the system? **1**
   a. -2^oC
   b. 0^oC
   c. 4^oC
   d. 2^oC
3. A brass boiler has a base area of 0.15 m^2 and thickness 1.0 cm. It boils water at the rate of 6.0 kg/min when placed on a gas stove. Estimate the temperature of the part of the flame in contact with the boiler. Thermal conductivity of brass = 109 J $s^{-1}m^{-1}K^{-1}$; Heat of vaporization of water = 2256 $\times 10^3 \mathrm{Jkg}^{-1}$. **1**
   a. 237 ^oC
   b. 238 ^oC
   c. 240 ^oC
   d. 239 ^oC
4. The Celsius temperature at absolute zero is equal to **1**
   a. 273^oC
   b. -273^oC
   c. -100^oC
   d. 100^oC
5. Two absolute scales A and B have triple points of water defined to be 200 A and 350 B. What is the relation between T_A and T_B ? **1**

a. T_A = (5/9) T_B
b. T_A = (4/7) T_B
c. T_A = (3/7) T_B
d. T_A = (2/7) T_B

6. A tightened glass stopper can be taken out easily by pouring hot water around the neck of the bottle. Why? **1**
7. These days people use steel utensils with copper bottom. This is supposed to be good for uniform heating of food. Explain this effect using the fact that copper is better conductor. **1**
8. What will be the effect of increasing temperature on (i) angle of contact (ii) surface tension? **1**
9. Find out the increase in moment of inertia I of a uniform rod (coefficient of linear expansion α) about its perpendicular bisector when its temperature is slightly increased by $\triangle$T. **2**



10. What do you mean by the term 'latent heat'? What are three types of latent heat? **2**
11. Find the temperature of 149°F on Kelvin scale. **2**
12. How does the density of a solid affected when the solid is heated? Derive the mathematical relation. **3**
13. What do you mean by absorptive power and reflecting power of a surface? What are their units? How are the two related to each other? **3**
14. Two rods of the same area of cross-section, but of lengths l_1 and l_2 and conductivities K_1 and K_2 are joined in series. Show that the combination is equivalent of a material of conductivity $K = \frac{l_1 + l_2}{\left(\frac{l_1}{K_1}\right)\left(\frac{l_2}{K_2}\right)}$ **3**
15. A hole is drilled in a copper sheet. The diameter of the hole is 4.24 cm at 27.0°C. What is the change in the diameter of the hole when the sheet is heated to 227°C? Coefficient of linear expansion of copper $= 1.70 \times 10^{-5} K^{-1}$. **5**